01 SEGMENTATION

WHAT HAPPENS IF YOU DIVIDE AN OBJECT?

1. divide the object into independent pieces
2. make the object easy to decompose
3. reinforce the degree of fragmentation

e.g., pizza in pieces, furniture from Ikea

02 SEPARATE

WHAT PART CAN YOU SEPARATE?

1. separate the part of the object or the property/function itself that is in question
2. separate out only the necessary part or property of the object

e.g., dietary supplements, perfume, contact lenses, franchise business, Google ads

03 LOCAL QUALITY

HOW COULD ANY PART OF THE FACILITY BE CHANGED?

1. change the structure of the facility or environment from homogeneous to heterogeneous
2. let each part of the facility function in the best conditions for itself
3. let each part of the object perform a different and useful function

e.g., a straw that can be bent, a perforation that makes it easier to tear a piece of paper, a date magnification in a watch, a magnetic tip in a screwdriver, a glove for touch screens

04 ASYMETRY

WHAT HAPPENS IF YOU CHANGE AN OBJECT ASYMMETRICALLY?

1. change the shape of the object from symmetrical to asymmetrical
2. if the object is asymmetrical, increase the asymmetry

e.g., Tripod, Boeing 737 engine air intake, Senz storm umbrella, asymmetrical earphone cable

05 CONNECTING

IS IT POSSIBLE TO COMBINE OBJECTS SO THAT THEY PERFORM OPERATIONS SIMULTANEOUSLY?

1. bring identical or similar objects closer or together, assemble similar/identical objects together so that they work together
2. let there be a combined or parallel operation, join them together in time

e.g., medical tour, all-in-one shampoo, hypermarket, charity concert, shopping Sunday

06 MULTIFUNCTIONALITY

IS IT POSSIBLE FOR AN OBJECT TO PERFORM MULTIPLE FUNCTIONS?

1. make the object or structure such that it performs more functions, make other parts redundant

e.g., Swiss Army pocketknife, smartphone, multifunctional fitness equipment, sofa that folds into a bed, two-way air conditioning

07 MATRYOSHKA - ONE INSIDE THE OTHER

WHAT HAPPENS IF YOU PUT ONE IN THE OTHER?

1. put one object into the other or put one object into the other one by one
2. let one part pass through the free space in the other

e.g., fireman's ladder, hamburger, ATM inside a bank, customized furniture

08 COUNTERWEIGHT

IS THERE ANY COUNTERWEIGHT TO OFFSET THE WEIGHT OF THE OBJECT?

1. to compensate for the weight of the object combine it with others to cause ascension
2. to compensate for the weight of the object let it interact with the environment

e.g., counterbalance for elevator cab, forklift, product placement as advertising

09 PREVIOUS COUNTERACTION

HOW CAN YOU TAKE A STEP BACK TO TAKE TWO STEPS FORWARD?

1. make an action with both harmful and useful effects
2. make a prior stress on the object

e.g., a vaccine, self-reporting to the police, a stand with free samples, a free trial period, a call for the return of products from a series in which a defect has been detected

10 PRIOR ACTION

WHAT CAN BE DONE BEFOREHAND?

1. make the necessary object change before it is needed (partially or completely)
2. arrange objects so that they are ready for action right away

e.g., an interim a return envelope, stickers on a phone that detect moisture, cut on a kitchen paper towel, a reservation.

11 ADVANCE COMPENSATION

HOW CAN YOU PREPARE FOR AN EMERGENCY SITUATION?

1. prepare for an accident - compensate in advance for the object's contingency

e.g. car airbag, spare wheel, fuse, insurance

12 LEVELING UP

CAN THE CONDITIONS OF OPERATION BE CHANGED TO MAKE OBJECTS HORIZONTALLY LEVEL?

1. change the conditions to eliminate the need to raise or lower objects in the gravity field

e.g., thicker shoe insole, movie credits, low-floor bus, children's menu in restaurant, height-adjustable chair

13 REVERSE

CAN IT BE DONE IN REVERSE?

1. do the activity in reverse
2. make the moving parts stationary and the non-moving parts moving
3. turn an object or process "upside down.,

e.g., library on wheels, homework, door-to-door sales, slow food

14 CURVES / ROUNDINGS

WHAT HAPPENS IF YOU USE CURVILINEAR OBJECTS?

1. instead of rectilinear parts, use curvilinear ones
2. instead of rectilinear motion, use rotary motion
3. use rollers, circles, spirals, domes, circularity

e.g., traffic circle, arches in architecture, rotary table, spiral stairs, revolving doors

15 DYNAMIZATION

WHICH PARTS CAN MOVE AGAINST EACH OTHER?

1. let the characteristics of the object find its optimal working conditions
2. If the object is rigid or inflexible, make it movable or flexible
3. divide the object into parts that can move relative to each other

e.g., a gimbal, a folding bicycle, extra seats in a car that are removed from the floor, a folding/rolling keyboard

16 PARTIALLY OR EXCESSIVELY

WHAT HAPPENS IF THERE IS A LITTLE LESS OR MORE OF SOMETHING?

1. if it is difficult to achieve 100% of an object by a certain way then use a little more or less of that way

e.g. buffet restaurant, post-it notes, special edition (limited), Beta version (underdeveloped)

17 CHANGE OF DIMENSION

WHAT WILL BE IN THE SPACE OF OTHER DIMENSIONS?

1. use an additional dimension in space
2. make a multilayer arrangement of objects instead of a single-layer arrangement
3. tilt the objects or reorient them spatially

e.g., multi-level parking lot, QR code, 3D movie, 3D booklet, Wall-mounted TV

18 VIBRATIONS

WHAT HAPPENS IF YOUR SUBJECT OBJECTS TO VIBRATION?

1. cause the object to oscillate or vibrate
2. increase the frequency of the vibration, such as to ultrasound
3. use the object's own resonant frequency

e.g. phone vibration, ultrasonic cleaning, ultrasound, ultrasonic humidifier

19 PERIODIC ACTION

WHAT HAPPENS IF PERIODIC ACTION IS USED?

1. instead of continuous action, use periodic or pulsating action
2. if the action is already periodic, change the amplitude or frequency
3. use pauses between pulses to perform another action

e.g., installment purchase, traffic regulation by traffic lights, automatic air freshener, lane shifted by the dispatcher to the needed direction

20 CONTINUOUS USEFUL ACTION

WHAT WILL BE DONE IF THE OPERATION IS CONTINUOUS?

1. conduct continuous operation. Let all parts work at maximum load
2. eliminate idle movements or intermittent operations
3. switch from reciprocating to rotating motion

e.g. 24-hour store, traffic circle, greenhouse, indoor ski slope, indoor stadium

21 ACCELERATE – JUMP

WHAT WILL HAPPEN TO OBJECTS AT HIGH SPEED?

1. run a process or certain steps at higher speeds to avoid what is harmful, destructive or risky

e.g., non-stop highway toll collection, drive-thru, UHT (seconds) pasteurization, high-speed freezing, ultra-fast cutting, RFID chips

22 TURN HARM INTO BENEFIT

IS IT POSSIBLE TO USE HARMFUL FACTORS TO ACHIEVE POSITIVE EFFECTS?

1. use harmful factors to achieve beneficial effects
2. eliminate the original harmful effect by combining it with another harmful effect
3. amplify the harmful factor to such an extent that it is no longer harmful

e.g. recovering waste materials, hiring hackers as security consultants, Botox

23 FEEDBACK

WHAT HAPPENS IF YOU ADD FEEDBACK?

1. introduce feedback
2. if the feedback already exists then change its strength or impact

e.g., evaluation survey, Facebook likes, article comments, checking questions on participants before including them in a scoring test, parking lot referral system, floats

24 INTERMEDIATE

WHAT INTERMEDIARY OBJECT CAN TEMPORARILY PERFORM THE ACTION?

1. use an intermediary or intermediate process
2. temporarily connect one object to another that can be easily removed

e.g., shoe spoon, medical capsules, chauffeur, real estate broker, absorbent for oil stain removal

25 SELF-SERVICE

HOW CAN YOU AUTOMATE?

1. let the facility operate itself by performing auxiliary operations
2. use waste or wasted resources, substances or energy

e.g., urinal sensor, self-assembling furniture, automatic faucet, self-service stand, self-driving robot vacuum cleaner, electronic banking

26 COPYING

ISN'T IT ENOUGH TO USE COPIES?

1. instead of inaccessible, expensive or fragile objects, use simpler and cheaper copies
2. turn the object into an optical copy

e.g., artificial flowers, mannequins at an exhibition, a scarecrow, a simulator, a video conference

27 PERISHABLE / NOT DURABLE BUT CHEAP

WHAT CAN BE REPLACED WITH DISPOSABLES?

1. replace an expensive object with many inexpensive objects by reducing the quality locally

e.g., disposable dishes or lenses, diapers, syringes

28 REPLACING MECHANICAL INTERACTION

WHAT OTHER SENSORY APPROACH CAN REPLACE THE MECHANICAL ONE?

1. replace mechanical means with sensory ones (optical, acoustic, taste, smell)
2. use fields - electric, mechanical or electromechanical to react with the object

e.g., books for the blind, an acoustic signal at traffic lights, a warning smell added to gas, an ultrasonic animal repellent

29 PNEUMO AND HYDRO STRUCTURES

CAN YOU USE GAS OR LIQUID INSTEAD OF SOLID PARTS?

1. use inflatable or liquid-filled parts

e.g., inflatable tent dome, waterbed, bubble wrap, inflatable collar/pillow for a long trip

30 FLEXIBLE COATINGS AND THIN FILMS

CAN YOU INSULATE/PROTECT AN OBJECT USING THIN FILMS?

1. use flexible sheaths and thin films instead of spatial structures
2. isolate the object from harmful external influences using flexible covers or thin films

e.g., air curtain, film covering the screen of a smartphone, UV filter cream, window tinting with a special film

31 POROSITY

WHAT HAPPENS IF YOU USE POROUS MATERIALS?

1. make the object porous or add porous elements
2. if the object is already porous, use the space in these pores for a useful purpose

e.g. insect netting, water filter, silicate gel (hygroscopic), foam soap

32 CHANGING OPTICAL PROPERTIES

IS IT POSSIBLE TO CHANGE THE COLOR OR TRANSPARENCY?

1. change the color of an object or its external environment
2. change the transparency of an object or its external environment

e.g. litmus paper, stickers that change color when exposed to temperature, transparent wrapping film, photochromic glasses, red stamp on Tefal pans, transparent button in an application

33 UNIFORMITY

WHAT HAPPENS IF THE OBJECTS ARE SIMILAR OR THE SAME?

1. let the objects react with a given object of the same material or with identical properties

e.g., a pair of wedding rings, a camouflage uniform, a street of cobblers, plates of bran (an edible bite), skin-colored bandages

34 REJECTION AND REGENERATION

IS IT POSSIBLE TO DISCARD OR REGENERATE AN OBJECT?

1. let the parts of the system that have already fulfilled their function drop off or modify them during operation
2. conversely, regenerate consumables during operation

e.g., multi-member rockets, absorbable surgical threads, refillable cartridges, second-hand stores, rechargeable batteries

35 PARAMETER CHANGE

IS IT POSSIBLE TO CHANGE THE PARAMETERS OF AN OBJECT?

1. change the physical state of the object (solid, gas, liquid)
2. change the consistency or concentration3. change the level of elasticity

e.g. popcorn, liquid soap, glue gun, liquid oxygen, wet wipes, LPG gas

36 PHASE TRANSITION

CAN YOU USE THE PHASE TRANSITION?

1. use the phenomena accompanying the phase transition (volume change, deformation, heat transfer)

e.g., air conditioning, sandblasting with dry ice, perfume, heating pipe, sprinkling salt on ice, steam engine

37 THERMAL EXPANSION

WHICH SPECIFIC PART DO YOU WANT TO EXPAND?

1. use the thermal expansion of materials
2. if you already use it, use the size of the multiplicity of materials with different properties

e.g. hot air balloon, mercury thermometer, bimetal, expansion spaces

38 ENHANCED INTERACTION

WHAT HAPPENS IF WE ADD SOMETHING REINFORCING?

1. instead of air, use oxygen
2. instead of oxygen, use ozone

e.g. catalyst, ozone sterilization, fermentation, cheerleaders, bonus or premium

39 NEUTRAL ENVIRONMENT

WHAT HAPPENS IF INERT PARTICLES ARE ADDED?

1. replace the environment with a neutral one
2. add something neutral or neutralizing additives to the object

e.g., aseptic room, nitrogen environment packaging, fire extinguisher, welcome page on Google, vacation

40 COMPOSITE MATERIALS

CAN NEW PROPERTIES BE OBTAINED BY JOINING OBJECTS?

1. change a homogeneous material into a composite (multi composite) structure

e.g., reinforced concrete, orchestra, bouquet of flowers, plywood, car window

40 TRIZ PRINCIPLES

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